ADR-002 - Syrka Domain Architecture, Bounded Contexts & System Ownership

Status: Accepted
Date: 2026-07-07
Builds on: ADR-001 - Selection of the Open-Source LMS Foundation for Syrka
Decision owners: Architecture / Engineering / Product
Scope: Permanent Syrka domain architecture, system ownership, data ownership, event contracts, and Open edX boundary

1. Executive Summary

ADR-001 selected an Open edX Hybrid approach: Open edX is a bounded operational LMS/runtime substrate, not the strategic product core. ADR-002 defines the permanent architecture that makes that decision safe.

Open edX is not Syrka. Open edX delivers learning experiences and captures educational activity. Syrka transforms those activities into human capability intelligence, career mobility, institutional analytics, workforce planning, and cross-border capability corridors.

The architectural philosophy is:

1. Syrka owns all intelligence. Ontologies, capability graphs, inference, career intent, market signals, national strategy alignment, recommendations, passports, and workforce mobility are Syrka-owned systems.
2. The LMS is replaceable. Open edX must remain behind an anti-corruption layer. Syrka must never depend directly on Open edX tables, internal APIs, event names, or implementation-specific identifiers.
3. Events are the backbone. Learning activity becomes canonical Syrka events. Canonical events become evidence. Evidence becomes capability claims. Capability claims become recommendations, credentials, job passports, and mobility decisions.
4. Each domain has one owner. No entity, table, event, or API may have shared ownership. Domains may reference each other by stable identifiers and consume each other’s events, but they do not mutate each other’s data.
5. AI decisions require provenance. No capability inference, opportunity recommendation, curriculum recommendation, or mobility judgment is accepted without evidence, confidence, explanation, model/version metadata, and bias/safety review.
6. Syrka protects its IP by owning the semantic layer. The proprietary value is not course delivery. It is the capability ontology, graph, inference engine, market alignment, career intelligence, passport, and corridor logic.

The permanent flow is:

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Student Learning
  v
Educational Evidence
  v
Canonical Syrka Event Bus
  v
Syrka Ontology
  v
Human Capability Graph
  v
Capability Inference Engine
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v
Career Intent Engine
v
Market Signal Engine
v
National Vision Layer
v
Syrka Odyssey
v
Capability Dashboard
v
Job Passport
v
Employer Platform
v
Government Platform
v
Workforce Mobility Engine
v
Cross-Border Capability Corridors

2. Domain-Driven Design

2.1 Domain boundaries

Syrka is a domain-driven platform composed of bounded contexts. Each bounded context owns a business capability, a domain model, persistence boundaries, APIs, events, and team accountability.

The primary bounded contexts are:

1. Learning Management
2. Identity & Access
3. Syrka Ontology
4. Human Capability Graph
5. Capability Inference Engine
6. Career Intent Engine
7. Market Signal Engine
8. National Vision Layer
9. Research Intelligence
10. GitHub Intelligence
11. LinkedIn Intelligence
12. Capability Dashboard
13. Portfolio Generator
14. Opportunity Engine
15. Employer Platform
16. Faculty Platform
17. Institutional Analytics
18. Curriculum Intelligence
19. Job Passport
20. Verified Credential Ledger
21. Workforce Mobility Engine
22. Syrka Corridors

23. Public Capability Profile
24. AI-Native Learning Analytics
25. Platform/Event Infrastructure

2.2 Context map

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flowchart LR
    LMS[Learning Management\nOpen edX bounded context] --> ACL[Learning Anti-Corruption Layer]
    ACL --> BUS[Canonical Syrka Event Bus]
    ID[Identity & Access] --> BUS
    BUS --> ONT[Syrka Ontology]
    BUS --> ANALYTICS[AI-Native Learning Analytics]
    ONT --> GRAPH[Human Capability Graph]
    BUS --> GRAPH
    GRAPH --> INF[Capability Inference Engine]
    INF --> DASH[Capability Dashboard]
    INF --> ODYSSEY[Syrka Odyssey]
    CAREER[Career Intent Engine] --> ODYSSEY
    MARKET[Market Signal Engine] --> OPPS[Opportunity Engine]
    NATIONAL[National Vision Layer] --> CURR[Curriculum Intelligence]
    RESEARCH[Research Intelligence] --> GRAPH
    GITHUB[GitHub Intelligence] --> GRAPH
    LINKEDIN[LinkedIn Intelligence] --> GRAPH
    OPPS --> EMP[Employer Platform]
    OPPS --> PASSPORT[Job Passport]
    CRED[Verified Credential Ledger] --> PASSPORT
    PASSPORT --> PUBLIC[Public Capability Profile]
    PASSPORT --> MOBILITY[Workforce Mobility Engine]
    MOBILITY --> CORRIDORS[Syrka Corridors]
    FACULTY[Faculty Platform] --> CURR
    INSTITUTION[Institutional Analytics] --> FACULTY
    ANALYTICS --> INSTITUTION
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2.3 Bounded context summary

Bounded context	Purpose	Data owner	Public APIs	Consumed events	Produced events	Integrations	Team owner
Learning Management	Deliver courses and assessments	Open edX bounded context	LMS adapter APIs only	IdentityProvisioned, EnrollmentRequested	LearningActivityObserved via ACL	Open edX, LTI tools	Learning Platform Team
Identity & Access	Unified identity, authn, authz, federation	Syrka Identity	Auth/session/user APIs	InstitutionCreated	IdentityCreated, RoleAssigned	Supabase Auth, SSO, OIDC, SAML	Platform Team
Syrka Ontology	Canonical semantic model of skills, knowledge, occupations, curricula	Syrka Ontology	Ontology query/admin APIs	LabourMarketSignalIngested, CurriculumMapped	OntologyTermCreated, OntologyVersionPublished	ESCO/O*NET/SOC/ISCO/custom taxonomies	Ontology Team
Human Capability Graph	Evidence-linked capability graph	Syrka Graph	Graph query/mutation APIs	CapabilityEvidenceObserved, OntologyVersionPublished	CapabilityGraphUpdated	Graph DB/vector store	Capability Intelligence Team
Capability Inference Engine	Infer capabilities from evidence	Syrka AI	Inference APIs	EvidenceObserved, GraphUpdated	CapabilityObserved, CapabilityUpdated	LLMs, ML models	AI Platform Team
Career Intent Engine	Model learner aspirations and constraints	Syrka Career	Intent APIs	StudentUpdated			

CareerIntentUpdated	Profile, dashboards	Student Intelligence Team
Market Signal Engine	Labour market ingestion and forecasting	Syrka Market	Market APIs	External market feeds	LabourMarketSignalIngested	Job boards, salary data, public data	Market Intelligence Team
National Vision Layer	Country strategy and workforce policy model	Syrka Government	Country/strategy APIs	GovernmentStrategyUpdated	NationalPriorityMapped	Ministries, public plans	Government Intelligence Team
Research Intelligence	Research output ingestion and mapping	Syrka Research	Research APIs	ResearchImported	ResearchCapabilityMapped	OpenAlex, Semantic Scholar, arXiv	Research Intelligence Team
GitHub Intelligence	Code/project evidence ingestion	Syrka Engineering Evidence	GitHub evidence APIs	GitHubRepositoryLinked	CodeCapabilityEvidenceObserved	GitHub	Evidence Integrations Team
LinkedIn Intelligence	Professional profile and signal ingestion	Syrka Professional Evidence	LinkedIn evidence APIs	LinkedInProfileLinked	ProfessionalSignalObserved	LinkedIn/exported data	Evidence Integrations Team
Capability Dashboard	Explain learner/institution capability state	Syrka Product	Dashboard APIs	CapabilityUpdated	DashboardViewed	Web apps	Product Experience Team
Portfolio Generator	Generate evidence-backed portfolios	Syrka Portfolio	Portfolio APIs	CapabilityUpdated, ProjectCompleted	PortfolioGenerated	Object storage, profiles	Product Experience Team
Opportunity Engine	Match people to jobs/projects/research/mobility	Syrka Opportunities	Recommendation APIs	CapabilityUpdated, CareerIntentUpdated, JobPosted	OpportunityMatched	Employers, markets	Opportunity Team
Employer Platform	Employer jobs, interest, verification	Syrka Employer	Employer/job APIs	OpportunityMatched	EmployerInterested, JobPosted	ATS/CRM	Employer Team
Faculty Platform	Faculty workflows and intervention tools	Syrka Faculty	Faculty APIs	LearningRiskDetected	FacultyInterventionCreated	LMS, dashboards	Academic Experience Team
Institutional Analytics	Institutional intelligence and benchmarking	Syrka Institution	Institution analytics APIs	LearningActivityObserved, CapabilityUpdated	InstitutionalInsightGenerated	BI/export	Institutional Team
Curriculum Intelligence	Map curriculum to skills/jobs/national priorities	Syrka Curriculum	Curriculum APIs	OntologyVersionPublished, LabourMarketSignalIngested	CurriculumGapDetected	LMS, faculty tools	Curriculum Team
Job Passport	Portable capability and credential artifact	Syrka Passport	Passport APIs	CredentialVerified, CapabilityUpdated	PassportIssued, PassportUpdated	Employers/governments	Passport Team
Verified Credential Ledger	Verifiable credential lifecycle	Syrka Credential	Credential APIs	AssessmentGraded, ExternalCredentialSubmitted	CredentialVerified, CredentialRevoked	VC/DID/blockchain optional	Trust Team
Workforce Mobility Engine	Determine mobility readiness and route	Syrka Mobility	Mobility APIs	PassportIssued, OpportunityMatched	MobilityApproved, MobilityBlocked	Governments/employers	Mobility Team
Syrka Corridors	Cross-border capability corridor orchestration	Syrka Corridors	Corridor APIs	MobilityApproved, NationalPriorityMapped	CorridorOpened, CorridorMatched	Governments/institutions	Corridor Team
Public Capability Profile	User-controlled public proof profile	Syrka Public Profile	Public profile APIs	PassportUpdated, PortfolioGenerated	PublicProfilePublished	Web, sharing, SEO	Public Platform Team
AI-Native Learning Analytics	Predict risk, mastery, engagement, quality	Syrka Analytics	Analytics APIs	LearningActivityObserved	LearningRiskDetected, MasterySignalDetected	LMS, dashboards	Analytics Team
Platform/Event Infrastructure	Event contracts, schemas, delivery, observability	Syrka Platform	Event/admin APIs	All domain events	EventAccepted, EventRejected	Kafka/NATS/Supabase queues	Platform Team

3. Complete Bounded Context Catalogue

3.1 Learning Management - Open edX bounded context

Purpose. Deliver structured learning experiences and collect raw educational activity.

Business responsibility. Open edX owns operational LMS workflows only: course authoring/runtime, modules, lessons, assignments, assessments, quizzes, discussions, attendance adapters, content delivery, gradebook, and enrolments. It does not own Syrka capability semantics.

Data ownership. Open edX owns LMS-local operational data. Syrka stores only canonical projections and evidence derived through the anti-corruption layer.

Internal models. Course, course run, unit, block, problem, assessment, submission, grade, discussion item, enrollment, cohort, content asset, learning sequence.

Public APIs. None directly exposed to strategic Syrka consumers. All access goes through learning-management-adapter APIs and canonical event streams.

Consumed events. IdentityProvisioned, EnrollmentRequested, CourseSyncRequested, ContentImportRequested.

Produced events. Open edX-native events are private. The anti-corruption layer emits canonical LearningActivityObserved, AssignmentSubmitted, AssignmentGraded, QuizCompleted, DiscussionParticipationObserved, AttendanceRecorded, and GradebookUpdated.

External integrations. LTI 1.3 tools, SCORM/xAPI bridge where needed, video/content storage, proctoring, attendance tools, AI tutors as LTI/XBlock tools.

Dependencies. Identity & Access for identity; Platform/Event Infrastructure for event publication; Object Storage for exported evidence artifacts.

Team ownership. Learning Platform Team.

What remains inside Open edX. Course content, course delivery, assessment delivery, LMS gradebook, discussion integration, enrolment mechanics, instructor LMS operations, and learning activity capture remain inside Open edX. Syrka does not place ontology, graph, inference, career, employer, passport, or mobility logic inside Open edX.

3.2 Identity & Access

Decision. Identity belongs to Syrka, implemented initially using Supabase Auth as the authentication provider. Open edX receives provisioned identities and SSO assertions; it is not the source of truth.

Purpose. Provide universal user identity across student, faculty, employer, institution, ministry, and public-profile surfaces.

Business responsibility. Authentication, authorization, tenant membership, institution federation, government federation, API access, service accounts, user lifecycle, consent, and audit.

Data ownership. Syrka Identity owns person identity, organization identity, tenant membership, roles, permissions, identity links, and consent records. Supabase Auth stores authentication credentials and sessions as an implementation component.

Internal models. User, person, organization, institution, government agency, employer, tenant, role, permission, policy, identity provider, federation link, consent grant, service account.

Public APIs. /identity/users, /identity/organizations, /identity/memberships, /identity/roles, /identity/consents, OIDC/SAML federation endpoints, token introspection.

Consumed events. InstitutionCreated, EmployerCreated, GovernmentAgencyCreated, UserInvited.

Produced events. StudentCreated, StudentUpdated, IdentityProvisioned, RoleAssigned, RoleRevoked, ConsentGranted, ConsentRevoked, FederationLinked.

External integrations. Supabase Auth, OIDC, SAML, university identity providers, government identity providers, employer SSO, Open edX SSO.

Dependencies. Platform/Event Infrastructure; Supabase PostgreSQL; Redis for sessions/rate limits.

Team ownership. Platform Team.

Authentication design. Supabase Auth handles passwordless, OAuth/OIDC, SAML via enterprise extensions or identity broker, MFA, and session tokens. Syrka issues domain authorization claims and short-lived service tokens.

Authorization design. Syrka uses policy-based RBAC/ABAC. RBAC grants coarse roles such as student, faculty, employer admin, ministry analyst, institutional admin, platform admin. ABAC constrains access by institution, country, consent, relationship, data sensitivity, and purpose.

Institution and government federation. Institutions and governments federate through OIDC/SAML. Syrka maps external groups to tenant-scoped roles; no external IdP may directly grant global Syrka privileges.

3.3 Syrka Ontology

Purpose. Define the canonical semantic model for human capability, knowledge, skills, occupations, curricula, credentials, industries, tools, and evidence types.

Business responsibility. Maintain versioned taxonomies and semantic relationships that all intelligence systems use.

Data ownership. Syrka Ontology owns terms, definitions, relationships, mappings, versions, deprecations, and alignment to external standards.

Internal models. OntologyTerm, Skill, KnowledgeArea, Capability, Occupation, Industry, Tool, CurriculumOutcome, CredentialType, EvidenceType, Relationship, Mapping, Version, JurisdictionProfile.

Public APIs. /ontology/terms, /ontology/skills, /ontology/capabilities, /ontology/occupations, /ontology/mappings, /ontology/versions, GraphQL semantic query API.

Consumed events. LabourMarketSignalIngested, CurriculumImported, CredentialFrameworkImported, NationalPriorityMapped.

Produced events. OntologyTermCreated, OntologyRelationshipCreated, OntologyVersionPublished, OntologyMappingDeprecated.

External integrations. ESCO, O*NET, SOC, ISCO, national qualification frameworks, university catalogs, professional bodies.

Dependencies. PostgreSQL for governance metadata; graph database for semantic relationships; vector database for semantic similarity.

Team ownership. Ontology Team.

Entity model. Terms are immutable within a version. Relationships include is_a, part_of, requires, enables, evidenced_by, maps_to, emerging_from, replaces, aligned_with, and regulated_by.

Taxonomy design. Syrka supports global base taxonomy plus country/institution overlays. Overlays may add local terms and mappings but may not mutate global term meaning.

Skill hierarchy. Skills decompose into capabilities, procedures, tools, knowledge prerequisites, proficiency levels, and evidence requirements.

Knowledge hierarchy. Knowledge areas map to concepts, topics, learning outcomes, evidence forms, and assessment patterns.

Occupation hierarchy. Occupations map to industries, roles, tasks, required capabilities, regulation constraints, salary bands, and market signals.

Curriculum hierarchy. Institutions map programs, courses, modules, outcomes, assessments, and rubrics to ontology terms.

3.4 Human Capability Graph

Purpose. Store the evolving, evidence-backed graph of what a person, cohort, institution, region, or workforce can credibly do.

Business responsibility. Maintain capability state, relationships, temporal evolution, evidence provenance, and confidence.

Data ownership. Entirely Syrka-owned. No LMS owns capabilities. An LMS only produces raw learning evidence.

Internal models. PersonNode, CapabilityNode, EvidenceNode, AssessmentNode, ProjectNode, CredentialNode, OccupationNode, InstitutionNode, EmployerNode, ConfidenceEdge, ProvenanceEdge, TemporalState.

Public APIs. /graph/person/{id}, /graph/capabilities, /graph/evidence, /graph/paths, /graph/cohorts, graph query API.

Consumed events. CapabilityEvidenceObserved, CapabilityObserved, CredentialVerified, ProjectCompleted, ResearchCapabilityMapped, CodeCapabilityEvidenceObserved.

Produced events. CapabilityGraphUpdated, CapabilityPathCreated, EvidenceAttached, CapabilityConfidenceChanged.

External integrations. Graph DB, vector DB, BI projections, privacy/consent service.

Dependencies. Ontology, Identity, Capability Inference Engine, Event Infrastructure.

Team ownership. Capability Intelligence Team.

Nodes. Person, capability, skill, knowledge concept, evidence item, assignment, project, repository, research output, credential, occupation, opportunity, institution, employer, jurisdiction.

Edges. has_evidence, demonstrates, requires, supports, contradicts, verified_by, derived_from, aligned_to, expires_at, observed_in, recommended_for.

Confidence. Confidence is stored as a first-class edge/property with score, model version, evidence count, evidence quality, recency, assessor credibility, and uncertainty.

Provenance. Every capability assertion points to source event IDs, evidence artifacts, evaluator identity/model, timestamp, jurisdiction, and consent basis.

Temporal relationships. Capability state changes over time. The graph stores observations and projections, never only the latest value.

Capability evolution. The graph supports progression, decay, reinforcement, transferability, specialization, and adjacent-skill inference.

3.5 Capability Inference Engine

Purpose. Convert evidence into capability observations, updates, explanations, and recommendations.

Business responsibility. Evaluate whether evidence supports specific capabilities and at what confidence/proficiency.

Data ownership. Owns inference runs, model outputs, explanations, evaluation metadata, bias checks, and audit records. It does not own source evidence or ontology terms.

Internal models. InferenceRun, EvidenceBundle, CapabilityHypothesis, ModelDecision, ConfidenceScore, Explanation, BiasCheck, HumanReview, EvaluationSet.

Public APIs. /inference/evaluate, /inference/explain/{runId}, /inference/models, /inference/review-queue.

Consumed events. LearningActivityObserved, AssignmentGraded, QuizCompleted, ProjectCompleted, ResearchPublished, GitHubRepositoryLinked, CredentialVerified, OntologyVersionPublished.

Produced events. CapabilityObserved, CapabilityUpdated, CapabilityRejected, InferenceReviewRequired, InferenceModelEvaluated.

External integrations. OpenAI/Claude-compatible LLMs, embedding models, rules engines, model registry, evaluation datasets.

Dependencies. Ontology, Human Capability Graph, Event Infrastructure, Vector DB.

Team ownership. AI Platform Team.

Inputs. Canonical events, assessment results, rubrics, artifacts, code repositories, research outputs, credential claims, market ontology, historical graph state.

Outputs. Capability observations, confidence deltas, proficiency estimates, gap assessments, explanations, human-review tasks, model evaluation metrics.

Algorithms. Rules for deterministic mappings, rubric-to-capability scoring, Bayesian confidence update, graph propagation, embedding similarity, supervised classifiers, LLM rubric reasoning with structured outputs, and ensemble adjudication.

Confidence scoring. Confidence combines evidence relevance, assessor quality, recency, difficulty, independence, consistency, authenticity, and model uncertainty.

Explainability. Every output includes evidence references, reasoning summary, ontology version, model version, counterfactual missing evidence, and limitations.

Bias mitigation. Protected attributes are excluded unless legally required for fairness measurement. Models are evaluated across cohorts, countries, languages, institutions, and accessibility contexts. High-impact decisions require human review and appeal.

3.6 Career Intent Engine

Purpose. Model what a learner wants to become and the constraints under which recommendations should be made.

Business responsibility. Capture aspirations, pathways, preferences, geographies, industries, salary expectations, risk tolerance, mobility intent, and life constraints.

Data ownership. Syrka Career owns career intent records and preference history.

Internal models. CareerIntent, Aspiration, Pathway, PreferredIndustry, PreferredLocation, SalaryExpectation, Constraint, MobilityPreference, WorkModePreference, TimeHorizon.

Public APIs. /career-intent, /career-pathways, /career-fit, /intent-history.

Consumed events. StudentCreated, CapabilityUpdated, OpportunityViewed, OpportunityAccepted, PassportIssued.

Produced events. CareerIntentUpdated, CareerPathwaySelected, CareerConstraintUpdated, MobilityIntentDeclared.

External integrations. Student profile UX, career advisors, labour market data, employer platform.

Dependencies. Identity, Ontology, Market Signal Engine, Human Capability Graph.

Team ownership. Student Intelligence Team.

3.7 Market Signal Engine

Purpose. Ingest and model labour market demand, salaries, skills, occupations, emerging sectors, and disruption risk.

Business responsibility. Convert market data into normalized, explainable, jurisdiction-aware market signals.

Data ownership. Syrka Market owns labour market signals, forecasts, source metadata, and demand models.

Internal models. JobPosting, SkillDemandSignal, SalaryBand, DemandForecast, EmergingIndustry, DisplacementRisk, RegionalLabourMarket, EmployerDemandProfile, SourceReliability.

Public APIs. /market/signals, /market/skills, /market/jobs, /market/salaries, /market/forecasts, /market/displacement.

Consumed events. JobPosted, EmployerInterested, NationalPriorityMapped, external ingestion events.

Produced events. LabourMarketSignalIngested, SkillDemandChanged, SalaryBandUpdated, EmergingIndustryDetected, AIDisplacementRiskUpdated.

External integrations. Job boards, government statistics, employer APIs, salary datasets, economic data, web crawlers where permitted.

Dependencies. Ontology, Event Infrastructure, Vector DB for job/skill matching.

Team ownership. Market Intelligence Team.

3.8 National Vision Layer

Purpose. Model country strategies, national development plans, strategic sectors, workforce policy, and capability priorities.

Business responsibility. Translate national strategies into computable priority models that influence curriculum intelligence and workforce planning.

Data ownership. Syrka Government owns country strategy profiles and priority mappings.

Internal models. CountryProfile, NationalStrategy, StrategicSector, WorkforceTarget, PolicyPriority, JurisdictionRule, CapabilityPriority, ProgramAlignment.

Public APIs. /national/countries, /national/strategies, /national/priorities, /national/workforce-targets.

Consumed events. GovernmentStrategyImported, LabourMarketSignalIngested, CurriculumGapDetected.

Produced events. NationalPriorityMapped, StrategicSectorUpdated, WorkforceTargetUpdated.

External integrations. Government portals, policy documents, ministries, international organizations.

Dependencies. Ontology, Market Signal Engine, Research Intelligence.

Team ownership. Government Intelligence Team.

3.9 Research Intelligence

Purpose. Transform research publications, projects, grants, labs, and collaborations into capability and opportunity signals.

Business responsibility. Ingest research metadata, map research themes to ontology, identify faculty/student capabilities, and surface collaboration opportunities.

Data ownership. Syrka Research owns normalized research records and research-capability mappings.

Internal models. Publication, Author, Affiliation, ResearchTopic, CitationSignal, Grant, Lab, CollaborationOpportunity, ResearchCapabilityMapping.

Public APIs. /research/publications, /research/topics, /research/collaborations, /research/capability-map.

Consumed events. ResearchImported, FacultyProfileUpdated, StudentProjectCompleted, OntologyVersionPublished.

Produced events. ResearchPublished, ResearchCapabilityMapped, ResearchCollaborationSuggested.

External integrations. OpenAlex, Semantic Scholar, arXiv, Crossref, ORCID, institutional repositories.

Dependencies. Ontology, Human Capability Graph, Vector DB.

Team ownership. Research Intelligence Team.

3.10 GitHub Intelligence

Purpose. Convert code repositories, commits, pull requests, issues, tests, and reviews into software capability evidence.

Business responsibility. Link user-controlled repositories, analyze contribution quality, and emit code capability evidence.

Data ownership. Syrka Engineering Evidence owns normalized GitHub evidence records, not GitHub's source data.

Internal models. RepositoryLink, CommitSignal, PullRequestSignal, ReviewSignal, TestSignal, LanguageProfile, CodeQualitySignal, ProjectEvidence.

Public APIs. /github/link, /github/repositories, /github/evidence, /github/capability-signals.

Consumed events. GitHubRepositoryLinked, ConsentGranted, ProjectCompleted.

Produced events. CodeCapabilityEvidenceObserved, RepositoryAnalyzed, GitHubRepositoryLinked, GitHubEvidenceRevoked.

External integrations. GitHub OAuth/API/webhooks, optional GitLab/Bitbucket adapters.

Dependencies. Identity, Consent, Ontology, Capability Inference Engine.

Team ownership. Evidence Integrations Team.

3.11 LinkedIn Intelligence

Purpose. Normalize user-consented professional profile signals into career and capability evidence.

Business responsibility. Import profile data, experience, endorsements, certifications, and role history where legally and contractually permitted.

Data ownership. Syrka Professional Evidence owns normalized professional signals and consent state.

Internal models. ProfessionalProfileLink, ExperienceSignal, CertificationSignal, EndorsementSignal, RoleHistory, OrganizationSignal.

Public APIs. /linkedin/link, /professional-profile/imports, /professional-signals.

Consumed events. ConsentGranted, LinkedInProfileLinked, CareerIntentUpdated.

Produced events. ProfessionalSignalObserved, LinkedInProfileLinked, ProfessionalSignalRevoked.

External integrations. LinkedIn APIs where available, user-uploaded exports, employer verification.

Dependencies. Identity, Consent, Ontology, Career Intent Engine.

Team ownership. Evidence Integrations Team.

3.12 Capability Dashboard

Purpose. Present capability state, evidence, confidence, gaps, and recommendations to students, faculty, institutions, employers, and governments.

Business responsibility. Explain what Syrka believes, why it believes it, how confident it is, and what action should happen next.

Data ownership. Owns dashboard layouts, saved views, user annotations, and insight presentation state. Does not own capability facts.

Internal models. DashboardView, CapabilityCard, EvidenceTimeline, GapInsight, RecommendationPanel, ExplanationView, SavedFilter.

Public APIs. /dashboards/capabilities, /dashboards/evidence, /dashboards/gaps, /dashboards/explanations.

Consumed events. CapabilityUpdated, CapabilityGraphUpdated, RecommendationGenerated, LearningRiskDetected.

Produced events. DashboardViewed, InsightAcknowledged, CapabilityDisputed.

External integrations. Web apps, PDF/export service, profile service.

Dependencies. Human Capability Graph, Inference Engine, Opportunity Engine, Identity.

Team ownership. Product Experience Team.

3.13 Portfolio Generator

Purpose. Generate human-readable and machine-verifiable portfolios from evidence-backed capabilities.

Business responsibility. Assemble projects, credentials, reflections, artifacts, and capability claims into audience-specific portfolios.

Data ownership. Owns portfolio documents, templates, exports, share links, and presentation metadata.

Internal models. Portfolio, PortfolioSection, EvidenceReference, Artifact, Template, ShareLink, AudienceProfile.

Public APIs. /portfolios, /portfolios/generate, /portfolios/share, /portfolios/export.

Consumed events. CapabilityUpdated, ProjectCompleted, CredentialVerified, PublicProfilePublished.

Produced events. PortfolioGenerated, PortfolioShared, PortfolioRevoked.

External integrations. Object storage, PDF generation, public profile, employer platform.

Dependencies. Human Capability Graph, Job Passport, Verified Credential Ledger.

Team ownership. Product Experience Team.

3.14 Opportunity Engine

Purpose. Match people to jobs, internships, research projects, scholarships, courses, mentors, and mobility routes.

Business responsibility. Produce explainable opportunity recommendations using capabilities, intent, market signals, and constraints.

Data ownership. Owns recommendation runs, match scores, recommendation explanations, and opportunity matching state. Opportunity entities are owned by their domain of origin.

Internal models. Opportunity, MatchRun, MatchScore, Recommendation, EligibilityRule, ConstraintViolation, Explanation, FeedbackSignal.

Public APIs. /opportunities, /opportunities/match, /recommendations, /recommendations/feedback.

Consumed events. CapabilityUpdated, CareerIntentUpdated, LabourMarketSignalIngested, JobPosted, ResearchCollaborationSuggested, MobilityApproved.

Produced events. OpportunityMatched, RecommendationGenerated, RecommendationAccepted, RecommendationRejected.

External integrations. Employer platform, faculty platform, market data, public profile.

Dependencies. Capability Graph, Career Intent, Market Signal, National Vision.

Team ownership. Opportunity Team.

3.15 Employer Platform

Purpose. Allow employers to publish demand, discover candidates, express interest, verify job-passport claims, and close feedback loops.

Business responsibility. Employer identity, jobs, projects, talent searches, candidate interest, hiring feedback, and demand validation.

Data ownership. Syrka Employer owns employer profiles, job postings, employer users, interest records, and hiring feedback.

Internal models. Employer, EmployerUser, Job, ProjectBrief, TalentSearch, EmployerInterest, HiringFeedback, VerificationRequest.

Public APIs. /employers, /employers/jobs, /employers/search, /employers/interests, /employers/verification.

Consumed events. PublicProfilePublished, OpportunityMatched, PassportIssued, CredentialVerified.

Produced events. EmployerCreated, JobPosted, EmployerInterested, HiringFeedbackSubmitted, VerificationRequested.

External integrations. ATS, CRM, email/calendar, government employment portals.

Dependencies. Identity, Job Passport, Opportunity Engine, Public Capability Profile.

Team ownership. Employer Team.

3.16 Faculty Platform

Purpose. Support faculty with capability-aware teaching, intervention, assessment design, and curriculum improvement.

Business responsibility. Faculty dashboards, cohort monitoring, intervention workflows, rubric insights, assessment feedback, and course-improvement loops.

Data ownership. Owns faculty-specific workflows, notes, interventions, recommendations, and teaching analytics views.

Internal models. FacultyView, CohortInsight, Intervention, RubricInsight, AssessmentQualitySignal, TeachingRecommendation.

Public APIs. /faculty/cohorts, /faculty/interventions, /faculty/rubrics, /faculty/recommendations.

Consumed events. LearningRiskDetected, MasterySignalDetected, CurriculumGapDetected, CapabilityUpdated.

Produced events. FacultyInterventionCreated, InterventionResolved, AssessmentImprovementRequested.

External integrations. LMS adapter, institutional systems, notification service.

Dependencies. Institutional Analytics, AI-Native Learning Analytics, Curriculum Intelligence.

Team ownership. Academic Experience Team.

3.17 Institutional Analytics

Purpose. Provide institution-level intelligence over capabilities, programs, cohorts, outcomes, equity, and market alignment.

Business responsibility. Aggregate institutional insights without owning underlying student capability facts.

Data ownership. Owns institutional metrics, aggregates, reports, benchmarks, and authorized exports.

Internal models. InstitutionMetric, CohortAggregate, ProgramBenchmark, EquityMetric, OutcomeMetric, AnalyticsReport, ExportJob.

Public APIs. /institutional/metrics, /institutional/cohorts, /institutional/reports, /institutional/exports.

Consumed events. LearningActivityObserved, CapabilityUpdated, CredentialVerified, OpportunityMatched, LearningRiskDetected.

Produced events. InstitutionalInsightGenerated, OutcomeBenchmarkUpdated, EquityRiskDetected.

External integrations. BI tools, accreditation reporting, ministry exports.

Dependencies. Identity, Analytics, Graph, Data Warehouse/PostgreSQL projections.

Team ownership. Institutional Team.

3.18 Curriculum Intelligence

Purpose. Evaluate and improve curricula by aligning courses, assessments, outcomes, capabilities, market signals, and national priorities.

Business responsibility. Detect gaps, duplication, outdated content, assessment misalignment, and strategic opportunities.

Data ownership. Owns curriculum maps, gap analyses, recommendations, and alignment reports. LMS owns course delivery; Curriculum Intelligence owns semantic curriculum understanding.

Internal models. CurriculumMap, CourseOutcomeMapping, AssessmentMapping, SkillCoverageMatrix, GapAnalysis, CurriculumRecommendation, AccreditationMapping.

Public APIs. /curriculum/maps, /curriculum/gaps, /curriculum/recommendations, /curriculum/alignment.

Consumed events. OntologyVersionPublished, CourseImported, AssessmentGraded, LabourMarketSignalIngested, NationalPriorityMapped.

Produced events. CurriculumGapDetected, CurriculumRecommendationGenerated, CourseOutcomeMapped.

External integrations. Open edX adapter, catalogs, accreditation bodies, ministry frameworks.

Dependencies. Ontology, Market Signal, National Vision, Learning Management adapter.

Team ownership. Curriculum Team.

3.19 Job Passport

Purpose. Produce a portable, evidence-backed representation of a person’s capabilities, credentials, preferences, and mobility readiness.

Business responsibility. Package verified capability claims for employers, governments, and cross-border mobility workflows.

Data ownership. Syrka Passport owns passport records, versions, shares, revocations, and disclosure policies.

Internal models. Passport, PassportVersion, CapabilityClaim, CredentialReference, MobilityReadiness, DisclosurePolicy, VerificationStatus, ShareToken.

Public APIs. /passports, /passports/issue, /passports/share, /passports/revoke, /passports/verify.

Consumed events. CapabilityUpdated, CredentialVerified, CareerIntentUpdated, MobilityIntentDeclared.

Produced events. PassportIssued, PassportUpdated, PassportShared, PassportRevoked.

External integrations. Employer platform, government platform, public profile, credential ledger.

Dependencies. Capability Graph, Credential Ledger, Identity, Consent.

Team ownership. Passport Team.

3.20 Verified Credential Ledger

Purpose. Manage credential claims, verification, revocation, issuer trust, and portable proof.

Business responsibility. Ensure credential assertions are verifiable, auditable, revocable, and tied to evidence.

Data ownership. Syrka Credential owns credential records, issuer registry, verification status, revocation state, and trust policies.

Internal models. Credential, CredentialClaim, Issuer, Verification, Revocation, TrustPolicy, EvidenceReference, VCExport.

Public APIs. /credentials, /credentials/verify, /credentials/issue, /credentials/revoke, /issuers.

Consumed events. AssessmentGraded, CredentialSubmitted, IssuerApproved, PassportIssued.

Produced events. CredentialVerified, CredentialIssued, CredentialRevoked, IssuerTrustChanged.

External integrations. W3C Verifiable Credentials, DID providers, credential issuers, optional blockchain anchoring.

Dependencies. Identity, Capability Graph, Object Storage, optional ledger infrastructure.

Team ownership. Trust Team.

3.21 Workforce Mobility Engine

Purpose. Determine whether a person, cohort, or talent pool is ready for internal, regional, or cross-border mobility.

Business responsibility. Match capability passports to mobility routes, visa/regulatory requirements, employer demand, government priorities, and readiness gaps.

Data ownership. Syrka Mobility owns mobility assessments, route decisions, readiness gaps, approvals, and case status.

Internal models. MobilityRoute, MobilityAssessment, ReadinessGap, EligibilityRule, RegulatoryRequirement, MobilityCase, ApprovalDecision.

Public APIs. /mobility/routes, /mobility/assess, /mobility/cases, /mobility/approvals.

Consumed events. PassportIssued, OpportunityMatched, CredentialVerified, NationalPriorityMapped, EmployerInterested.

Produced events. MobilityApproved, MobilityBlocked, MobilityGapDetected, MobilityCaseOpened.

External integrations. Government agencies, immigration/regulatory datasets, employer platform, corridor service.

Dependencies. Passport, National Vision, Market Signal, Credential Ledger.

Team ownership. Mobility Team.

3.22 Syrka Corridors

Purpose. Operate cross-border capability corridors between countries, institutions, employers, and sectors.

Business responsibility. Coordinate demand, supply, standards, credentials, mobility rules, and reporting across jurisdictions.

Data ownership. Syrka Corridors owns corridor definitions, agreements, eligibility mappings, pipeline status, and corridor analytics.

Internal models. Corridor, CorridorPartner, CorridorAgreement, CapabilityStandard, CandidatePipeline, CorridorMatch, JurisdictionMapping.

Public APIs. /corridors, /corridors/partners, /corridors/pipelines, /corridors/matches, /corridors/reports.

Consumed events. MobilityApproved, NationalPriorityMapped, WorkforceTargetUpdated, EmployerInterested.

Produced events. CorridorOpened, CorridorMatched, CorridorCapacityUpdated, CorridorOutcomeReported.

External integrations. Governments, institutions, employers, credential authorities.

Dependencies. Mobility Engine, National Vision, Employer Platform, Credential Ledger.

Team ownership. Corridor Team.

3.23 Public Capability Profile

Purpose. Provide user-controlled public profiles that expose selected capability evidence, portfolios, passports, and credentials.

Business responsibility. Manage public presentation, privacy, consent, sharing, SEO, verification views, and revocation.

Data ownership. Syrka Public Profile owns profile pages, slugs, visibility settings, shared claims, and access logs.

Internal models. PublicProfile, ProfileSection, VisibilityPolicy, SharedClaim, VerificationView, AccessLog.

Public APIs. /profiles/public, /profiles/{slug}, /profiles/share, /profiles/visibility.

Consumed events. PassportUpdated, PortfolioGenerated, CredentialVerified, ConsentRevoked.

Produced events. PublicProfilePublished, PublicProfileViewed, PublicProfileRevoked.

External integrations. Web, social sharing, employer verification, search indexing where allowed.

Dependencies. Identity, Portfolio, Passport, Consent.

Team ownership. Public Platform Team.

3.24 AI-Native Learning Analytics

Purpose. Analyze learning activity for mastery, risk, engagement, assessment quality, and pedagogical effectiveness.

Business responsibility. Produce learning analytics signals from canonical events while respecting privacy and explainability.

Data ownership. Syrka Analytics owns analytics models, aggregates, risk signals, mastery signals, and explainability metadata.

Internal models. LearningSession, EngagementSignal, MasterySignal, RiskPrediction, AssessmentQualityMetric, CohortPattern, AnalyticsModelRun.

Public APIs. /analytics/learning, /analytics/mastery, /analytics/risk, /analytics/quality, /analytics/models.

Consumed events. LearningActivityObserved, AssignmentSubmitted, AssignmentGraded, QuizCompleted, DiscussionParticipationObserved, AttendanceRecorded.

Produced events. LearningRiskDetected, MasterySignalDetected, AssessmentQualitySignalGenerated, EngagementChanged.

External integrations. Faculty platform, institutional analytics, capability inference engine.

Dependencies. Event Infrastructure, Learning Management adapter, Graph, Inference Engine.

Team ownership. Analytics Team.

4. Data Ownership Matrix

No entity may have shared ownership. Other domains may hold read models, projections, references, or cached views only.

Entity	Single owner	Storage of record	Notes
Student	Syrka Identity	Supabase PostgreSQL/Auth	Open edX receives provisioned LMS account only
Person profile	Syrka Identity	Supabase PostgreSQL	Public profile is a separate presentation entity
Institution	Syrka Identity	Supabase PostgreSQL	Institution analytics consumes it
Government agency	Syrka Identity	Supabase PostgreSQL	National Vision references it
Employer	Employer Platform	Supabase PostgreSQL	Identity owns employer users/memberships
Course	Learning Management	Open edX DB	Curriculum Intelligence stores mappings only
Module/lesson	Learning Management	Open edX DB	Canonical IDs are mapped in ACL
Assignment	Learning Management	Open edX DB	Syrka stores evidence projection only
Assessment/quiz	Learning Management	Open edX DB	Results become canonical events
Grade	Learning Management	Open edX DB	Syrka stores evidence and derived capability claims
Enrollment	Learning Management	Open edX DB plus Syrka projection	Source of LMS enrollment remains Open edX
Skill	Syrka Ontology	Ontology DB/Graph	Versioned semantic object
Knowledge concept	Syrka Ontology	Ontology DB/Graph	Versioned semantic object
Capability	Human Capability Graph	Graph DB	References ontology terms
Capability observation	Capability Inference Engine	PostgreSQL + Graph projection	Graph consumes accepted observations
Evidence item	Human Capability Graph	Graph DB + object storage refs	Source artifact remains in source system
Career intent	Career Intent Engine	Supabase PostgreSQL	User-controlled and versioned
Research output	Research Intelligence	PostgreSQL/object refs/vector	Mapped to graph as evidence
Project	Portfolio Generator or source domain	Supabase PostgreSQL/object storage	Learning project from LMS remains LMS-owned until normalized as evidence
GitHub repository link	GitHub Intelligence	Supabase PostgreSQL	GitHub owns repository; Syrka owns normalized evidence
LinkedIn profile link	LinkedIn Intelligence	Supabase PostgreSQL	User consent required
Credential	Verified Credential Ledger	Credential DB/ledger	Passport references verified credentials
Job	Employer Platform	Supabase PostgreSQL	Opportunity Engine recommends jobs
Opportunity match	Opportunity Engine	Supabase PostgreSQL	Does not own underlying opportunity entity
Passport	Job Passport	Supabase PostgreSQL/object storage	Versioned, revocable
Public capability profile	Public Capability Profile	Supabase PostgreSQL/object storage	User-controlled disclosure

Government strategy	National Vision Layer	PostgreSQL/vector/object refs	Parsed from official sources
Labour market signal	Market Signal Engine	PostgreSQL/vector/time-series	Versioned by source and timestamp
Curriculum map	Curriculum Intelligence	Supabase PostgreSQL/Graph	Does not own LMS course content
Institutional report	Institutional Analytics	Warehouse/PostgreSQL/object storage	Aggregate projection only
Mobility case	Workforce Mobility Engine	Supabase PostgreSQL	References passport/opportunity
Corridor	Syrka Corridors	Supabase PostgreSQL	Cross-jurisdiction orchestration
Event schema	Platform/Event Infrastructure	Schema registry	Canonical contract owner

5. Canonical Event Architecture

5.1 Universal event envelope

Every canonical event uses the same envelope:


```
{
  "event_id": "uuid",
  "event_type": "AssignmentSubmitted",
  "event_version": "1.0.0",
  "occurred_at": "2026-07-07T00:00:00Z",
  "published_at": "2026-07-07T00:00:01Z",
  "producer": "learning-management-adapter",
  "source_system": "openedx",
  "tenant_id": "uuid",
  "country_code": "SA",
  "institution_id": "uuid",
  "subject": { "type": "student", "id": "uuid" },
  "correlation_id": "uuid",
  "causation_id": "uuid",
  "idempotency_key": "source-system:source-event-id:version",
  "privacy": { "classification": "restricted", "consent_basis": "education" },
  "payload": {},
  "provenance": { "source_event_id": "...", "source_url": "...", "adapter_version": "..." },
  "schema_url": "https://schemas.syrka.ai/events/AssignmentSubmitted/1.0.0"
}
```


5.2 Event catalogue

Event	Producer	Primary consumers	Payload summary	Retention/replay
StudentCreated	Identity	LMS adapter, Graph, Dashboards	student_id, tenant, institution, role metadata	Permanent identity audit; replay to provision services
StudentUpdated	Identity	Career, Dashboards, Analytics	changed profile fields, tenant scope	Permanent audit with PII controls
AssignmentSubmitted	Learning ACL	Analytics, Inference, Graph	assignment_id, attempt_id, artifact refs, timestamps	Long-term educational evidence; replayable
AssignmentGraded	Learning ACL	Inference, Graph, Faculty, Analytics	grade, rubric, feedback, assessor, attempt	Long-term evidence; replayable
QuizCompleted	Learning ACL	Analytics, Inference	quiz_id, score, attempts, duration, item-level summary	Long-term evidence with sensitive detail controls
ProjectCompleted	Learning ACL/Portfolio/GitHub	Graph, Portfolio, Inference	project_id, artifacts,	

collaborators, rubric | Long-term portfolio/evidence |
ResearchPublished	Research Intelligence	Graph, Portfolio, Opportunity	publication metadata, authors, topics	Permanent scholarly evidence
GitHubRepositoryLinked	GitHub Intelligence	Inference, Graph	repo link, consent, ownership proof	Retained while consent active
CapabilityObserved	Inference Engine	Graph, Dashboard, Passport	capability_id, evidence refs, confidence, explanation	Permanent with model provenance
CapabilityUpdated	Graph	Dashboard, Opportunity, Passport	capability state delta, confidence, timestamp	Permanent graph history
CareerIntentUpdated	Career Intent	Opportunity, Mobility, Odyssey	aspirations, preferences, constraints	User-controlled history
RecommendationGenerated	Opportunity/Odyssey	Dashboard, Audit	recommendation, score, explanation, model	Retain for audit and learning
PassportIssued	Job Passport	Employer, Mobility, Public Profile	passport_id, version, claims, disclosure policy	Permanent/revocable versions
CredentialVerified	Credential Ledger	Graph, Passport, Employer	credential_id, issuer, verification result	Permanent trust audit
OpportunityMatched	Opportunity Engine	Student, Employer, Mobility	opportunity_id, match score, reasons	Retain for audit and feedback
EmployerInterested	Employer Platform	Opportunity, Mobility, Analytics	employer_id, profile/passport target, job	Retain under consent and policy
MobilityApproved	Mobility Engine	Corridors, Passport, Government dashboards	route, eligibility, conditions, expiry	Permanent case history

5.3 Versioning

- Event names are stable business facts, not implementation names.
- Schema versions use semantic versioning.
- Consumers must tolerate additive fields.
- Breaking changes create a new major version and dual-publish during migration.
- A schema registry stores JSON Schema/OpenAPI/AsyncAPI definitions.

5.4 Retry and idempotency

- Producers attach deterministic idempotency_key values.
- Event ingestion is at-least-once; consumers must be idempotent.
- Failed deliveries go to dead-letter queues with replay tooling.
- Consumer checkpoints are stored per consumer group.

5.5 Security

- Events are signed by producers or delivered through authenticated infrastructure.
- Event payloads are classified: public, internal, restricted, highly restricted.
- PII is minimized; sensitive artifacts are referenced by secure object IDs rather than embedded.
- Domain authorization applies to event subscriptions.

5.6 Retention and replay

- Canonical events are immutable.
- Long-term evidence events are retained for the lifetime required by product, legal, and institutional policy.
- Retention is domain-specific and country-aware.
- Replay can rebuild graph projections, dashboards, analytics aggregates, and passports.

6. Anti-Corruption Layer

The Learning Anti-Corruption Layer is the permanent translation boundary between Open edX and Syrka.

```
flowchart LR
    OEX[Open edX native DB/APIs/events] --> EXTRACT[Extractors]
    EXTRACT --> MAP[Mapping Registry]
    MAP --> VALIDATE[Validation]
    VALIDATE --> TRANSFORM[Transformation]
    TRANSFORM --> CANON[Canonical Event Builder]
    CANON --> BUS[Syrka Event Bus]
    VALIDATE --> DLQ[Dead Letter Queue]
    DLQ --> OPS[Ops Review & Replay]
```

Mapping. Open edX courses, blocks, submissions, grades, discussions, and enrollment records are mapped to Syrka canonical identifiers using a mapping registry. The registry stores source_system, source_id, canonical_id, entity_type, valid_from, and valid_to.

Validation. The ACL validates source completeness, tenant ownership, consent/policy basis, schema compatibility, timestamp sanity, duplicate status, and required evidence references.

Transformation. The ACL converts Open edX concepts into canonical Syrka facts. Example: an Open edX problem attempt becomes QuizCompleted or AssessmentAttempted, not an Open edX-specific event.

Error handling. Invalid records are rejected into a dead-letter queue with source pointer, validation errors, adapter version, and replay eligibility. Poison messages are quarantined.

Monitoring. Metrics include extraction lag, transform error rate, event publish success, unmapped entity rate, duplicate rate, DLQ volume, and per-tenant throughput.

Logging. Logs must not contain student submissions or PII beyond stable IDs. Sensitive payloads are stored in secure object storage and referenced.

Versioning. Each adapter has a version. Mapping changes are migration-controlled. Source version and adapter version are written into event provenance.

Migration strategy. If Open edX is replaced, only this ACL changes. Canonical event contracts, ontology, graph, dashboards, inference, passport, and mobility systems remain unchanged.

7. API Ownership

Domain	Public APIs	Internal APIs	Protocols	AuthN/AuthZ	Rate limiting	Consumers	Provider/owner
Learning Management Adapter	course projections, enrollment requests, evidence exports	Open edX connector APIs	REST/events	Syrka service tokens + RBAC	tenant/service limits	Web, Curriculum, Analytics	Learning Platform Team
Identity	users, orgs, roles, consents, federation	policy evaluation, token introspection	REST/OIDC/SAML	Supabase Auth + Syrka policies	per user/org/IP	all domains	Platform Team
Ontology	terms, mappings, versions	ontology admin/versioning	REST/GraphQL	service/user policies	per		

tenant and API key | graph, inference, curriculum | Ontology Team |
Graph	capability graph, evidence, paths	graph mutation/event consumers	GraphQL/REST/events	consent + purpose policy	strict per purpose	dashboards, passport, opportunity	Capability Intelligence Team
Inference	evaluate, explain, review	model adapters, batch jobs	REST/events/workers	service identity + policy	cost/model budgets	graph, analytics	AI Platform Team
Career Intent	intent, pathways, constraints	scoring features	REST/events	user-owned + advisor scopes	per user	opportunity, mobility	Student Intelligence Team
Market	signals, salaries, forecasts	ingestion jobs	REST/events	service/user policies	per org/API key	opportunity, curriculum, national	Market Intelligence Team
National Vision	country strategies, priorities	policy ingestion	REST/events	government/institution scopes	per tenant	curriculum, mobility	Government Intelligence Team
Research	publications, topics, collaborations	source ingestion	REST/events	institution/user scopes	per source	graph, portfolio	Research Team
Evidence Integrations	GitHub/LinkedIn links and signals	webhook handlers	REST/webhooks/events	OAuth + consent	provider-specific	inference, graph	Evidence Integrations Team
Dashboard	capability views and explanations	saved-view service	REST/GraphQL	user/tenant roles	per user	web apps	Product Experience Team
Portfolio	portfolio generate/share/export	template renderer	REST/events	user consent	per user/export	public profile, employer	Product Experience Team
Opportunity	matches/recommendations	scorer workers	REST/events	user/employer roles	per user/org	web, employer, mobility	Opportunity Team
Employer	employers/jobs/search/interests	ATS connectors	REST/webhooks	employer RBAC	per employer	opportunity, passport	Employer Team
Faculty	cohort/intervention/rubric APIs	LMS/faculty projections	REST/events	faculty/institution roles	per institution	faculty app	Academic Experience Team
Institutional Analytics	metrics/reports/exports	aggregate builders	REST/events	institution roles	export quotas	admin/ministry	Institutional Team
Curriculum	maps/gaps/recommendations	mapping workers	REST/events	faculty/admin roles	per institution	faculty, ministry	Curriculum Team
Passport	issue/share/verify	claim builder	REST/events	user consent + verifier scopes	per verifier	employer, mobility	Passport Team
Credential Ledger	issue/verify/revoke	issuer registry	REST/events/VC	issuer trust policy	per issuer	passport, employer	Trust Team
Mobility	routes/assess/cases	rule engine	REST/events	government/employer/user scopes	case quotas	corridors	Mobility Team
Corridors	corridors/pipelines/reports	partner integrations	REST/events	corridor partner roles	per partner	governments	Corridor Team
Public Profile	publish/share/view	rendering/cache	REST	public + signed URLs	public abuse limits	web/employers	Public Platform Team
Analytics	mastery/risk/quality	model workers	REST/events	institution/faculty scopes	per tenant	faculty/institution	Analytics Team

REST is the default for transactional APIs. GraphQL is appropriate for graph/dashboard read composition. gRPC is reserved for high-throughput internal model or scoring calls. Events are required for domain state propagation.

8. Database Ownership

8.1 Supabase PostgreSQL

Purpose. Primary application relational store for Syrka-owned transactional domains, tenant models, API projections, audit metadata, and operational read models.

Owner. Platform Team owns the cluster and standards; each domain owns its schema.

Schema strategy. Use schema-per-domain: identity, ontology, career, market, national, portfolio, opportunity, employer, faculty, institutional, curriculum, passport, credentials, mobility, corridors, public_profile, analytics, events.

Indexes. All tenant-scoped tables index tenant_id; all event projections index event_id, occurred_at, subject_id; all user-owned tables index user_id plus consent_state where relevant.

Relationships. Cross-domain references use stable IDs only. No cross-domain foreign keys that create deployment coupling.

Partitioning. Event projections and analytics aggregates partition by tenant/country/time depending on query patterns.

Scaling. Read replicas for dashboards; logical replication into warehouse; row-level security for user/tenant access; connection pooling.

8.2 Graph Database

Purpose. Capability graph, ontology relationships, evidence-to-capability relationships, occupation paths, and mobility routes.

Owner. Capability Intelligence Team for capability graph; Ontology Team for ontology subgraph. Platform owns infrastructure standards.

Data. Nodes and edges described in sections 3.3 and 3.4.

Indexes. Node IDs, ontology term IDs, person IDs, capability IDs, temporal validity, relationship types, confidence ranges.

Scaling. Partition by tenant/person/country as needed; materialize read models for common dashboard queries.

8.3 Vector Database

Purpose. Semantic retrieval over evidence artifacts, course descriptions, resumes, publications, job postings, national strategies, and ontology terms.

Owner. AI Platform Team, with domain-owned collections.

Collections. evidence_embeddings, job_embeddings, research_embeddings, curriculum_embeddings, policy_embeddings, ontology_embeddings, portfolio_embeddings.

Indexes. ANN indexes by collection; metadata filters for tenant, country, language, source, consent, and freshness.

Scaling. Store embeddings with model/version metadata. Re-embed on model upgrades with dual-index migration.

8.4 Redis

Purpose. Caching, rate limiting, short-lived sessions, job locks, idempotency windows, and queue coordination.

Owner. Platform Team.

Data. No durable domain state. All Redis data must be rebuildable.

8.5 Object Storage

Purpose. Store artifacts: submissions, portfolios, PDFs, research files, credential proofs, profile media, exports, and evidence snapshots.

Owner. Platform Team infrastructure; domains own buckets/prefixes and retention policy.

Security. Signed URLs, encryption, classification metadata, virus/malware scanning, retention tags, legal hold where required.

8.6 Open edX database

Purpose. LMS-local operational persistence only.

Owner. Learning Platform Team.

Rule. Syrka services may not query Open edX databases directly. Only the ACL may read/translate Open edX state.

9. Service Architecture

Bounded context	Service form	Justification
Learning Management	External bounded platform + adapter service	Open edX remains isolated and replaceable
Identity & Access	Modular platform service	Shared dependency with strong security boundary
Syrka Ontology	Modular service + graph-backed admin	Versioned domain used by many services
Human Capability Graph	Dedicated service + event consumer	Central strategic graph with specialized persistence
Capability Inference Engine	AI service + background workers	Model orchestration, batch scoring, human review
Career Intent Engine	Modular service	User-owned profile/intention workflows
Market Signal Engine	Ingestion workers + API service	External feeds, forecasts, scheduled jobs
National Vision Layer	Ingestion workers + API service	Policy document parsing and government workflows
Research Intelligence	Ingestion workers + API service	External APIs and semantic mapping
GitHub Intelligence	Integration service + webhook workers	Provider OAuth/webhooks and code analysis
LinkedIn Intelligence	Integration service + ingestion workers	Consent-heavy professional signal ingestion
Capability Dashboard	Web/BFF module	Presentation composition over graph/inference APIs
Portfolio Generator	Background worker + API service	Document generation and object storage exports
Opportunity Engine	Recommendation AI service + API	Scoring, ranking, feedback learning
Employer Platform	Modular service + web app	Employer workflows and job lifecycle
Faculty Platform	Web/BFF module + workflow service	Institution/faculty UX and interventions
Institutional Analytics	Event consumers + aggregate API	Heavy read models and reports
Curriculum Intelligence	AI service + mapping workers	Semantic curriculum analysis and recommendations
Job Passport	Dedicated service	Trust, versioning, sharing, revocation
Verified Credential Ledger	Dedicated trust service	Security-critical credential lifecycle
Workforce Mobility Engine	Rule engine + workflow service	Eligibility, routes, cases, approvals
Syrka Corridors	Workflow/orchestration service	Multi-party cross-border operations

Public Capability Profile	Public web service + cache	Public traffic, privacy, SEO/security boundary
AI-Native Learning Analytics	Event consumers + AI service	Predictive analytics over canonical learning events
Platform/Event Infrastructure	Shared platform services	Event bus, schema registry, observability

10. Repository Architecture

Recommended future monorepo layout:

```
apps/
  web/                 # student and public Syrka web app
  admin/               # platform administration
  employer/            # employer portal
  faculty/             # faculty workflows
  ministry/            # government/ministry platform
  public-profile/      # public capability profile surface

services/
  learning-management-adapter/
  identity/
  ontology/
  capability-graph/
  capability-inference/
  career-intent/
  market-signals/
  national-vision/
  research-intelligence/
  github-intelligence/
  linkedin-intelligence/
  opportunity-engine/
  portfolio-generator/
  employer-platform/
  faculty-platform/
  institutional-analytics/
  curriculum-intelligence/
  job-passport/
  credential-ledger/
  workforce-mobility/
  syrka-corridors/
  learning-analytics/

packages/
  ui/                  # shared design system
  auth/                # client/server auth helpers
  sdk/                 # generated API clients
  events/              # canonical event types, schemas, publishers
  database/            # DB clients and migration helpers
  types/               # shared TypeScript types
  telemetry/           # logging, tracing, metrics helpers
  policy/              # authorization policy helpers
  ai/                  # model client abstractions, not business logic
  testing/             # test utilities and fixtures
```

```
infrastructure/
  terraform/
  docker/
  k8s/
  observability/
  event-bus/
  secrets/

supabase/
  migrations/
  seed/
  policies/
  functions/

docs/
  adr/
  architecture/
  schemas/
  runbooks/
```

Ownership rules.

- apps/* are owned by Product Experience teams but may not contain domain business logic beyond presentation orchestration.
- services/* are owned by the named domain teams.
- packages/events is owned by Platform/Event Infrastructure and reviewed by affected domain owners.
- packages/ai owns model-provider abstractions only; inference rules belong in domain services.
- supabase/migrations are schema-per-domain and reviewed by the owning domain team plus Platform.
- infrastructure is owned by Platform/SRE.

11. Technology Ownership

Technology	Owning domain/use	Why
Next.js	User-facing apps and BFF routes	Fits current Syrka web architecture and modern UX
TypeScript	Apps, API surfaces, event contracts	Strong contracts and Claude-assisted maintainability
Supabase Auth	Identity implementation provider	Fast, secure auth foundation while Syrka owns identity model
Supabase PostgreSQL	Transactional app persistence	Strong relational source for Syrka-owned data
PostgreSQL RLS	Identity/tenant security	Enforces row-level tenant and user boundaries
Redis	Platform cache/session/rate limits	Ephemeral performance and coordination
Graph Database	Ontology and capability graph	Natural fit for semantic capability relationships
Vector Database	AI retrieval and semantic matching	Required for LLM grounding, similarity, recommendations
Open edX	Learning Management bounded context	Course/runtime/assessment foundation selected in ADR-001
Python	AI, inference, ingestion, graph workflows	Best ecosystem for ML/LLM/data pipelines
TypeScript workers	Event adapters and product-adjacent jobs	Shared types and integration speed
Claude/OpenAI-compatible LLMs	AI services through provider abstraction	Model vendors are replaceable; prompts/evals are Syrka IP
Object Storage	Evidence artifacts, exports, passports	Durable binary/object evidence storage

Event Bus	Platform/Event Infrastructure	Backbone for replaceability and domain decoupling
Mermaid/ADR docs	Architecture governance	Makes decisions reviewable and enforceable

12. Replaceability Analysis

12.1 If Open edX disappears tomorrow

Services that continue operating: Identity, Ontology, Capability Graph, Inference over existing evidence, Career Intent, Market Signal, National Vision, Research Intelligence, GitHub Intelligence, LinkedIn Intelligence, Dashboards over existing projections, Portfolio, Opportunity, Employer Platform, Institutional Analytics over existing projections, Curriculum Intelligence over existing mappings, Job Passport, Credential Ledger, Mobility, Corridors, Public Profiles, and AI-Native Analytics over historical events.

Services that fail or degrade: New LMS course delivery, new LMS assignment submissions, new LMS quizzes, LMS gradebook sync, LMS-native discussions, attendance capture tied to Open edX, and any Open edX-specific authoring workflow.

Why Syrka survives. Canonical events, graph state, passports, credentials, and intelligence services are not stored in Open edX. Historical evidence remains replayable from Syrka event storage and object storage.

12.2 Replacing Open edX with Canvas

Required work. Build a Canvas ACL adapter, map Canvas courses/assignments/submissions/grades/users to canonical IDs, integrate Canvas SSO, validate LTI behavior, replay backfills, and dual-run during migration.

Estimated effort. 3-6 months for parity with core LMS activity events; 6-12 months for deep feature parity, depending on institution workflows.

12.3 Replacing Open edX with Moodle

Required work. Build Moodle ACL adapter, map Moodle plugin/activity events, implement SSO, migrate course structures where required, map gradebook and activity modules, validate event coverage.

Estimated effort. 4-8 months for core event parity; 9-15 months for broad plugin-heavy deployments.

12.4 Replacing Open edX with custom LMS

Required work. Build course runtime, content authoring, assessments, gradebook, enrolments, discussions/attendance integrations, LTI support, and canonical event producer natively.

Estimated effort. 12-24 months for credible institutional LMS parity; less if the custom LMS is intentionally narrow and Syrka controls all content formats.

13. Intellectual Property Boundary

Syrka’s proprietary IP is concentrated in:

1. Syrka Ontology - the semantic model connecting education, capability, occupations, markets, and national strategies.

2. Human Capability Graph - the evidence-backed representation of human potential and verified capability evolution.
3. Capability Inference Engine - algorithms, prompts, evaluations, confidence models, explainability, and bias controls.
4. Career Intent Engine - aspiration and pathway intelligence.
5. Market Signal Engine - normalized demand, salary, emerging-industry, and AI-displacement signals.
6. National Vision Layer - computable strategy alignment for governments.
7. Opportunity Engine - matching and recommendation logic.
8. Job Passport - portable, trustworthy, evidence-backed capability artifact.
9. Verified Credential Ledger - credential trust, verification, revocation, and issuer logic.
10. Workforce Mobility Engine and Syrka Corridors - cross-border capability mobility rules and orchestration.
11. AI-native Learning Analytics - predictive and explanatory learning intelligence.
12. Canonical Event Schemas and Anti-Corruption Layer - the abstraction that makes every LMS replaceable.

These components must remain independent of LMS implementation details. Open edX, Canvas, Moodle, or any future LMS may feed the system; none may define Syrka's core semantics.

14. Architecture Principles

1. LMSs are replaceable. No strategic Syrka domain may depend on LMS-specific schemas or APIs.
2. Syrka owns all intelligence. Capabilities, inference, graphs, recommendations, passports, and mobility decisions are Syrka-owned.
3. Events are immutable. Corrections are new events, not mutations of history.
4. Every capability must have evidence. Capability claims without evidence are drafts or hypotheses, never verified facts.
5. No AI decision without provenance. Model outputs require evidence references, model version, confidence, explanation, and audit record.
6. Every domain has one owner. Shared ownership is forbidden. Read models and references are allowed.
7. APIs are contracts. Domains integrate through versioned APIs/events, not database access.
8. Domain models are independent of implementation. Supabase, Open edX, graph DBs, and LLM providers are implementation choices.
9. Consent is a product primitive. User-controlled data sharing must be explicit, revocable, and auditable.
10. Privacy is designed in. Minimize data, classify payloads, and separate artifacts from event metadata.
11. Security follows the data. Authorization is based on user, tenant, role, purpose, consent, and jurisdiction.
12. Country independence is mandatory. Country overlays may specialize policy, but global architecture remains stable.
13. Institution independence is mandatory. Institutions configure mappings and policies; they do not fork Syrka semantics.
14. AI vendors are replaceable. Prompt templates, evaluations, schemas, and provenance are Syrka IP; providers are adapters.
15. Reproducibility matters. Inference runs must be replayable or explainably non-replayable when models are deprecated.
16. Observability is required. Every event, API, worker, model run, and adapter must emit logs, metrics, and traces.
17. Human review exists for high-impact decisions. Mobility, credential disputes, and adverse opportunity decisions require appeal paths.
18. Data lineage is non-negotiable. Every derived object must point to source events and

evidence artifacts.
19. Open standards are preferred. Use OIDC, SAML, LTI, W3C VC, OpenAPI, AsyncAPI, and portable data formats where possible.
20. Architecture is governed by ADRs. Future deviations require new ADRs that explicitly update or supersede this constitution.

15. Architecture Diagrams

15.1 System Context Diagram

```
flowchart TB
    Student[Student] --> Syrka[Syrka Platform]
    Faculty[Faculty] --> Syrka
    Employer[Employer] --> Syrka
    Institution[Institution Admin] --> Syrka
    Government[Government / Ministry] --> Syrka
    Syrka --> LMS[Open edX LMS bounded context]
    Syrka --> Market[Labour market sources]
    Syrka --> Research[Research sources]
    Syrka --> GitHub[GitHub]
    Syrka --> LinkedIn[LinkedIn / professional data]
    Syrka --> LLM[LLM and AI providers]
    Syrka --> Credential[Credential issuers/verifiers]
```

15.2 Bounded Context Map

```
flowchart LR
    Learning[Learning Management] --> ACL[Anti-Corruption Layer]
    ACL --> Events[Canonical Event Bus]
    Identity[Identity & Access] --> Events
    Events --> Ontology[Syrka Ontology]
    Ontology --> Graph[Human Capability Graph]
    Events --> Graph
    Graph --> Inference[Capability Inference]
    Inference --> Career[Career Intent]
    Market[Market Signal] --> Opportunity[Opportunity Engine]
    Career --> Opportunity
    Graph --> Opportunity
    National[National Vision] --> Curriculum[Curriculum Intelligence]
    Curriculum --> Faculty[Faculty Platform]
    Graph --> Passport[Job Passport]
    Credentials[Credential Ledger] --> Passport
    Passport --> Mobility[Workforce Mobility]
    Mobility --> Corridors[Syrka Corridors]
    Opportunity --> Employer[Employer Platform]
```

15.3 Container Diagram

```
flowchart TB
```

```
  subgraph Apps
    Web[Next.js Student Web]
    FacultyApp[Next.js Faculty App]
    EmployerApp[Next.js Employer App]
    MinistryApp[Next.js Ministry App]
    PublicApp[Public Profile App]
  end
  subgraph Platform
    API[API Gateway / BFF]
    Auth[Identity Service + Supabase Auth]
    Bus[Event Bus + Schema Registry]
    Obs[Observability]
  end
  subgraph Learning
    OpenEdX[Open edX]
    Adapter[Learning Adapter / ACL]
  end
  subgraph Intelligence
    Ont[Ontology Service]
    GraphSvc[Capability Graph Service]
    Infer[Inference Service]
    Analytics[Learning Analytics]
    Opp[Opportunity Engine]
  end
  subgraph TrustMobility
    Passport[Job Passport]
    Cred[Credential Ledger]
    Mobility[Mobility Engine]
    Corridors[Corridors]
  end
  Web --> API
  FacultyApp --> API
  EmployerApp --> API
  MinistryApp --> API
  PublicApp --> API
  API --> Auth
  OpenEdX --> Adapter
  Adapter --> Bus
  Bus --> Ont
  Bus --> GraphSvc
  Bus --> Analytics
  Ont --> GraphSvc
  GraphSvc --> Infer
  Infer --> Opp
  GraphSvc --> Passport
  Cred --> Passport
  Passport --> Mobility
  Mobility --> Corridors
  Obs -. monitors .- API
  Obs -. monitors .- Bus
```

15.4 Component Diagram - Capability Inference

```
flowchart LR
    Events[Canonical Events] --> Evidence[Evidence Bundler]
    Evidence --> Retriever[Vector/Graph Retriever]
    Retriever --> Rules[Deterministic Rules]
    Retriever --> LLM[LLM Rubric Reasoner]
    Rules --> Ensemble[Decision Ensemble]
    LLM --> Ensemble
    Ensemble --> Confidence[Confidence Scorer]
    Confidence --> Explain[Explanation Builder]
    Explain --> Review{Human review required?}
    Review -- yes --> Queue[Review Queue]
    Review -- no --> Publish[Publish CapabilityObserved]
```

15.5 Event Flow Diagram

```
sequenceDiagram
    participant LMS as Open edX
    participant ACL as Learning ACL
    participant Bus as Syrka Event Bus
    participant Analytics as Learning Analytics
    participant Infer as Capability Inference
    participant Graph as Capability Graph
    participant Dash as Capability Dashboard
    LMS->>ACL: Native submission/grade event
    ACL->>ACL: Map + validate + transform
    ACL->>Bus: AssignmentSubmitted v1
    Bus->>Analytics: Deliver event
    Bus->>Infer: Deliver event
    Infer->>Graph: Read evidence/context
    Infer->>Bus: CapabilityObserved v1
    Bus->>Graph: Update capability edge
    Graph->>Bus: CapabilityUpdated v1
    Bus->>Dash: Refresh user view
```

15.6 Data Flow Diagram

```
flowchart TB
    LMS[Open edX activity] --> ACL[ACL transformation]
    ACL --> EventStore[(Immutable Event Store)]
    EventStore --> Projections[(Supabase PostgreSQL projections)]
    EventStore --> Graph[(Capability Graph DB)]
    EventStore --> Vector[(Vector DB)]
    Artifacts[Evidence artifacts] --> Object[(Object Storage)]
    Object --> Vector
    Graph --> Dashboards[Dashboards]
    Vector --> Inference[Inference Engine]
    Graph --> Inference
    Inference --> EventStore
    Projections --> Apps[Next.js Apps]
```

15.7 Sequence Diagram - Replacing Open edX

```
sequenceDiagram
    participant Canvas as New LMS Adapter
    participant Registry as Mapping Registry
    participant Bus as Canonical Event Bus
    participant Graph as Capability Graph
    participant Passport as Job Passport
    Canvas->>Registry: Register Canvas source IDs to canonical IDs
    Canvas->>Bus: Publish AssignmentSubmitted v1
    Bus->>Graph: Existing consumer receives unchanged event
    Graph->>Bus: CapabilityUpdated v1
    Bus->>Passport: Existing consumer receives unchanged event
    Note over Graph,Passport: No domain service changes because canonical contract is stable
```

15.8 Deployment Diagram

```
flowchart TB
    subgraph Edge
        CDN[CDN/WAF]
    end
    subgraph Vercel
        Apps[Next.js Apps]
        BFF[API Routes / BFF]
    end
    subgraph CloudPlatform
        Services[Domain Services]
        Workers[Background Workers]
        Bus[Event Bus]
        Redis[(Redis)]
        Obs[Logs/Metrics/Traces]
    end
    subgraph Data
        PG[(Supabase PostgreSQL)]
        GraphDB[(Graph DB)]
        VectorDB[(Vector DB)]
        ObjectStore[(Object Storage)]
    end
    subgraph LearningRuntime
        OpenEdX[Open edX Deployment]
        Adapter[Learning ACL Adapter]
    end
    CDN --> Apps
    Apps --> BFF
    BFF --> Services
    Services --> PG
    Services --> GraphDB
    Services --> VectorDB
    Services --> ObjectStore
```

```
Services --> Redis
Services --> Bus
Workers --> Bus
OpenEdX --> Adapter
Adapter --> Bus
Obs -. monitors .- Services
Obs -. monitors .- Workers
Obs -. monitors .- Adapter
```

16. Final Decision

Syrka's permanent architecture is a domain-driven, event-driven, AI-native platform in which Open edX is only the Learning Management bounded context. Open edX stops at course delivery, assessment operations, gradebook mechanics, enrolments, discussions, attendance capture, and raw educational activity.

Syrka begins at canonical event ownership, ontology, graph, inference, career intent, market intelligence, national vision alignment, dashboards, portfolios, opportunities, passports, credentials, mobility, public profiles, and cross-border corridors.

All future engineering work must preserve the replaceability of Open edX. Any proposal that places Syrka ontology, capability inference, graph state, job passport logic, employer intelligence, workforce mobility, or national strategy logic inside Open edX violates this ADR unless superseded by a later accepted ADR.